Ankit Sinha

AI Infrastructure / Backend Engineer | Reliable Systems | Open Source
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Education

Maharaja Agrasen Institute of Technology — New Delhi
B.Tech in Information Technology — CGPA: 7.69/10.0 — Sept 2023 – June 2027

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
AI Infrastructure: LLM Gateways, Cost Budgeting, Provider Fallback, Usage Metering
Backend: Node.js, Express.js, FastAPI, REST APIs, Zod, Pydantic, JWT, AuthN/AuthZ
Data & Infrastructure: PostgreSQL, Redis, Drizzle ORM, Supabase, Docker Compose, Render, BullMQ
Reliability & Testing: Transactions, Row Locks, Idempotency, Transactional Outbox, Append-only Ledger, Retries, Circuit Breakers, Rate Limiting, Cancellation, pytest, Vitest, Integration Testing, CI/CD (GitHub Actions), Git

Open-Source Contributions

Agent Swarm - Merged PRs #2754 & #2755 — TypeScript, Bun
- Rejected credential-bearing feedback URLs and blocked 0.0.0.0/8 hosts (including IPv4-mapped IPv6) in browser reachability checks; added regression tests.

Faker.js - Merged PR #4020, v11.0 Milestone — TypeScript, Vitest
- Refactored romanNumeral tests to use public inputs instead of internal spies.

Agenta - Merged PR #6453 — Python, pytest
- Fixed schemeless Python SDK API URLs to prevent malformed OTLP endpoints; added 7 unit tests.

Projects

LLM Gateway — Budget-Controlled LLM API Gateway — GitHub — Live
TypeScript, Node.js, Express.js, PostgreSQL, Drizzle ORM, Groq, Gemini, Render
- Built virtual-key authentication and per-key usage accounting; kept provider credentials server-side.
- Reserved estimated request cost in integer micro-USD using PostgreSQL row locks; committed before LLM calls and settled provider-reported usage afterward.
- Added timeouts, transient retries, Groq-to-Gemini fallback and a primary-provider circuit breaker; propagated client disconnects to cancel upstream calls and release budget reservations.
- Added TTL recovery for abandoned reservations; wrote 11 tests with mock providers covering the TOCTOU reservation race, budget exhaustion, fallback, and usage accounting.

GateKeep — Encrypted Content-Sharing App — GitHub — Live
TypeScript, Node.js, Express.js, PostgreSQL, Drizzle ORM, Supabase Storage, Docker Compose, Render
- Built JWT-authenticated note sharing and private uploads with expiring, passphrase-protected links.
- Enforced atomic read limits and burn-on-read semantics to prevent concurrent over-consumption.
- Encrypted notes with AES-256-GCM; protected Supabase uploads with short-lived signed URLs.
- Implemented HMAC-signed webhooks with retries; added 24 unit and 8 PostgreSQL-backed integration tests (32 total).

IdleForge — Idle-Game Economy Backend — GitHub
TypeScript, Node.js, Express.js, PostgreSQL, Drizzle ORM, Redis, BullMQ, Docker Compose
- Built atomic offline collection and transactional upgrades with guarded SQL and row locks against double credit or spend.
- Added persisted idempotency for client retries plus a transactional outbox and PostgreSQL occurrence records for scheduled bonuses, so lost or duplicate jobs never pay twice.
- Applied hourly bonuses in resumable batches with a persisted player cursor; wrote append-only ledger entries in every economy transaction and added a `ledger:verify` invariant check.
- Built a PostgreSQL-reconciled Redis sorted-set leaderboard; returned explicit 503 responses when leaderboard data was unavailable instead of serving misleading empty results.
- Added 72 tests, including 46 integration tests on real PostgreSQL/Redis; a 300-request mixed load test at 10 concurrency returned 0 errors at p95 28.5 ms and 519 req/s.

Achievements & Recognition

Winner — GenAI Track, Code Kshetra 2.0 — Led backend development for the winning application.
Pre-Finalist — BVP HEX Hackathon and HackWithMAIT 6.0.